

## CS 528: Unconditional Lower Bounds for Algorithms and Data Structures

Course Code: CS 528

Course Name: Unconditional Lower Bounds for Algorithms and Data Structures

L-T-P-C: 3-0-0-3

Prerequisites: CS 208, CS 212 or equivalent

Intended for: M.Tech / Ph.D. / B.Tech 3rd and 4th year

Mutual Exclusion: None

Approval: 64th BoA (proposed as CS 612)

### Preamble

The theory of NP-completeness allows us to prove that a computational problem cannot have polynomial-time algorithms (i.e. a superpolynomial lower bound on the time required to solve the problem). However, such lower bounds rely on the unproven assumption that  $P \neq NP$ . In this course, we will study some techniques for proving unconditional lower bounds for many computational problems.

### Course Modules

- **Unit 1: Motivation via streaming algorithms:** data stream model, estimation of frequency moments, connection with one-way communication. (5 hours)
- **Unit 2: One-way communication complexity:** definition of one-way protocols, definition of disjointness, index, and gap-hamming problem, lower bound for disjointness and index. (6 hours)
- **Unit 3: Lower bounds for streaming algorithms:** space lower bound for  $F_\infty$  estimation, reducing (one-way communication) gap-hamming problem to  $F_0$  estimation, lower bound on the one-way communication complexity of gap-hamming. (8 hours)
- **Unit 4: Two-way communication complexity:** deterministic and randomized protocols, monochromatic rectangles, fooling sets, distribution complexity, corruption method. (12 hours)
- **Unit 5: Lower bounds for data structures:** the approximate nearest neighbour problem, cell-probe model, asymmetric communication complexity, lower bound for the approximate nearest neighbour problem. (11 hours)

### Laboratory/Practical/Tutorial Modules

None

### Text Books

1. Tim Roughgarden, *Communication Complexity (for Algorithm Designers)*, Now Publishers, 2016.

### References

1. Amir Yehudayoff and Anup Rao, *Communication Complexity and Applications*, Cambridge University Press, 2020.



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2. Eyal Kushilevitz and Noam Nisan, *Communication Complexity*, Cambridge University Press, 2009.